



Research on Deep Learning Method of Crop Disease Identification

Yin Long

Wuhan Polytechnic University
P.O. Box 430040
Wuhan, China
875919201@qq.com

Changhua Liu*

Wuhan Polytechnic University
P.O. Box 430040
Wuhan, China
liuch@whpu.edu.cn

ABSTRACT

Traditional image processing has many problems in crop disease recognition, such as complicated and inefficient manual design. This paper studies the performance of deep learning algorithms in crop diseases. The research mainly optimizes the convolutional neural network from two aspects, one is the model structure, and the other is training label optimization. The basic convolutional neural network uses ResNet-50, InceptionV3, MobileNetV2, etc. Model training uses transfer learning to compare experimental data based on the performance of each model. the Top-1 accuracy rate of ResNet-B reached 91.2%, and the Top-5 accuracy rate was as high as 99.8%. Top1 accuracy of other models is increased by at least 0.7%

CCS CONCEPTS

• Computing methodologies • Artificial intelligence • Computer vision • Computer vision problems • Object identification

KEYWORDS

Image recognition, Convolutional neural network, Crop disease, Model optimization

1 Introduction

Crop disease identification is the key technology for crop pest and disease protection. With the advancement of science and technology, the development of agriculture has been inseparable from image processing technology. Traditional crop disease identification has problems such as long detection time and complicated process [1]. With the rapid development of the deep learning method [2] in the field of image recognition, the

combination of deep learning methods and crop diseases has become the trend of research in the field.

Before combining deep learning techniques, crop disease research methods mainly focus on image segmentation, feature extraction and classification and recognition [3]. For example, in 2002, Xu Guili extracted the color characteristics of tomato disease color pictures through histogram [4]. In 2009, Qi Guangyun successfully combined the genetic algorithm and neural network to identify the lesion image of soybean leaves, and the accuracy rate reached 90% [5]. In 2012, He Dandan proposed a algorithm for rice leaf edge detection based on multi-strategy fusion technology combined with median filtering and Canny operator to identify rice diseases [6]. The methods of image segmentation are mostly based on the specific laboratory environment and there are still great differences in practical applications. The method of feature extraction class has a single method for extracting features, and the degree of feature analysis and optimization is not enough. Most of the algorithms for classification and recognition focus on feature engineering, but this method is complicated and inefficient, and the symptoms of the same crop disease are different at different times, which greatly increases the workload of the classification recognition algorithm.

With the increase of crop disease image data, it provides the most important foundation for the application of deep learning technology to crop disease identification. Mohanty S P [7] used 54,306 public data sets on crop diseases in 2016, which included diseased and healthy plant leaves. By training deep convolutional neural networks to identify 14 crops and 26 diseases, the algorithm has an accuracy of more than 90% on the test set. Amanda R [8] used the cassava disease image dataset collected in the wild in Tanzania in 2017. The training convolutional neural network successfully identified three diseases and the types of damage caused by the two pests, with the best performing model up to 93%.

The datasets in this paper are classified in terms of disease severity. The convolutional neural network structures used in training are VGG-16 [9], ResNet-50 [10], and InceptionV3 [11], MobileNet [12]. The research mainly optimizes the convolutional neural network from two aspects, one is the model structure, and the other is training label optimization. The model training adopts the idea of transfer learning [13]. During the training

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process, the hyperparameters of the model are continuously adjusted, and the optimization algorithm is used to finally complete the crop disease identification and compare the performance of each algorithm.

2 Data Processing and Experimental Environment

2.1 Introduction to the Data Set

The data source is AI Challenger (<https://challenger.ai>), which distinguishes 61 categories according to the degree of disease of the species, including 10 species and 27 diseases. After analyzing the data of the image dataset, it was decided to delete the 44 and 45 categories. After processing, the experimental crop disease data set has 40,772 images. The data set has 59 categories, with 10 crops (*apples*, *tomatoes*, *cherries*, *strawberries*, etc.) and 26 disease samples. There are 23 types of diseases classified into general and severe according to the degree of disease. Image labels are labeled with plant type, disease type, and disease severity. The training set is about 70%, the verification set is about 15%, and the test set is about 15%. Figure 1 shows the disease picture of the data set:

2.2 Data Preprocessing

Data is enhanced before the image is input into the convolutional neural network model. This experiment uses online enhancement to obtain the batch data, and then enhance the batch data, such as: random horizontal flip of the image, random vertical flip of the image, image The center is the center of rotation, press $(-30 \sim +30)$ to rotate the image, and so on. At the same time, the image is resized and the picture RGB three-channel pixels are normalized to $[-1, +1]$. Image normalization enables similar distribution of all images, which makes it easier to converge during training, speeds up convergence, and reduces model training time. Data augmentation In this experiment, not only the training data is increased, but also the generalization

ability of the model is improved, and the robustness of the model is improved by adding noise data. Suppose the pixel of the image is $M = (r, g, b)$.

$$M' = (r / 255, g / 255, b / 255) \quad (1)$$

From (1), the pixel range of the image can be $[0, 1]$, $mean = [0.48, 0.457, 0.406]$ $std = [0.229, 0.224, 0.225]$,

$$M_e = \frac{M' - mean}{std} \quad (2)$$

From (2), it can be calculated that the final value range of the pixel point M is in $[-1, 1]$, mean and std are mean and variance, and the value is calculated and processed from the training set.

2.3 Experimental Environment

The software environment of this experiment is based on the python 3.6 and the pytorch0.4.1 deep learning framework to construct a convolutional neural network model. Hardware environment: The operating system is Win10 64-bit, the processor AMD Ryzen 5 2600 six-core, 16G memory, and the graphics card is Nvidia GeForce GTX 1060TI 6GB memory.

3 Model Training Optimization

3.1 Transfer Learning

Wei Chao, deep learning-based crop disease detection [14], this article discusses the model performance of the transfer learning and non-transfer learning methods, and the experimental data set in this article is also provided by AI Challenger. The experimental results are better with the transfer learning model. Therefore, this experiment uses the technique of migration learning. The model initialization parameters are predefined parameters of the convolutional network such as MobileNet on ImageNet, and the final output neurons of the model are changed to 59 species of crop diseases.

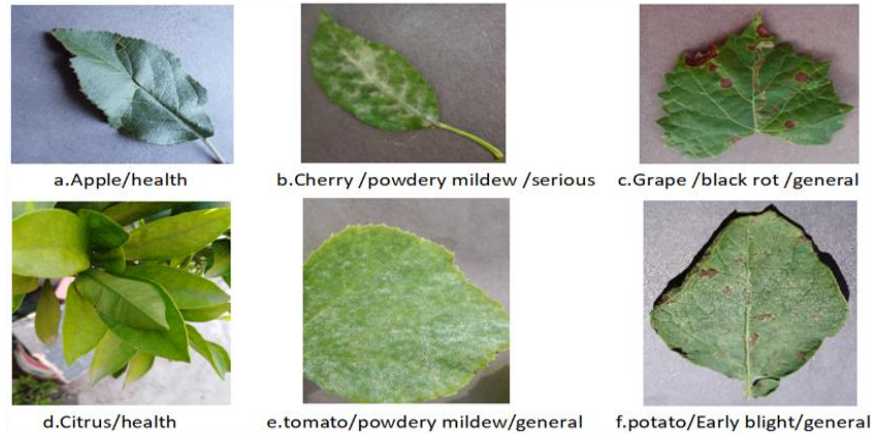


Figure 1: Part of the Crop Disease Picture in the Data Set.

3.2 Binary Cross Entropy Loss Function

In this experiment, the binary cross entropy loss function is used. The cross entropy describes the distance between two probability distributions. The smaller the cross entropy, the smaller the distance between the two. However, the output of the neural network is not necessarily a probability distribution, so we use the Softmax regression to turn the result of the neural network forward propagation into a probability distribution. Assuming that the output of the neural network is: $A_1, A_2 \dots A_n$, then the output after softmax regression processing is as follows (3):

$$\text{soft max}(A)_i = y'_i = \frac{e^{A_i}}{\sum_{j=1}^n e^{A_j}} \quad (3)$$

As can be seen from the above Formula (3), the new output satisfies all the requirements of the probability distribution. y'_i For the output after softmax processing, assume that the correct label for this output is: $y_1, y_2 \dots y_n$, Then the loss function of the binary cross entropy is as follows (4):

$$\text{Cross Entropy Loss} = -\sum_{i=1}^n y_i \log y'_i \quad (4)$$

For image classification problems, cross entropy is generally used as the loss function. In the actual training process, L2 regularization is usually used to suppress over-fitting. The final loss function of the model is (5):

$$\text{Loss} = -\sum_{i=1}^n y_i \log y'_i + \lambda \sum_j w_j^2 \quad (5)$$

3.3 Optimization Algorithm

In this experiment, the algorithm flow of Adam (Adaotive Moment Estimation) algorithm [15] is:

Initial time: $V_{dw} = 0, S_{dw} = 0, V_{db} = 0, S_{db} = 0$; On iteration t:

Compute dW,db;

$$V_{dw} = \beta_1 V_{dw} + (1 - \beta_1) dw, V_{db} = \beta_1 V_{db} + (1 - \beta_1) db \quad (6)$$

$$V_{dw}^{\text{Corrected}} = \frac{V_{dw}}{1 - \beta_1^t}, V_{db}^{\text{Corrected}} = \frac{V_{db}}{1 - \beta_1^t} \quad (7)$$

$$S_{dw} = \beta_2 S_{dw} + (1 - \beta_2) dw^2, S_{db} = \beta_2 S_{db} + (1 - \beta_2) db^2 \quad (8)$$

$$S_{dw}^{\text{Corrected}} = \frac{S_{dw}}{1 - \beta_2^t}, S_{db}^{\text{Corrected}} = \frac{S_{db}}{1 - \beta_2^t} \quad (9)$$

$$W := W - \partial \frac{V_{dw}^{\text{Corrected}}}{\sqrt{S_{dw}^{\text{Corrected}} + \epsilon}}, b := b - \partial \frac{V_{db}^{\text{Corrected}}}{\sqrt{S_{db}^{\text{Corrected}} + \epsilon}} \quad (10)$$

The above V_{dw} and V_{db} are the gradients of W and b calculated by the Momentum gradient algorithm. S_{dw} and S_{db}

are the gradients of W and b calculated by the RMSprop algorithm. Because the initial setting is 0, the gradient value of w and b at the beginning is small, so offset correction is used to solve this problem. $V_{dw}^{\text{Corrected}}, V_{db}^{\text{Corrected}}, S_{dw}^{\text{Corrected}}, S_{db}^{\text{Corrected}}$

Is the gradient after offset correction. Where t is the number of iterations of training, $\alpha, \beta_1, \beta_2, \epsilon$ is the hyperparameter of the Adam algorithm, where ∂ is the learning factor, β_1, β_2 is the gradient average and the coefficient of the square, and ϵ is to avoid the denominator of the RMSprop algorithm expression is zero, so this pole is added Small constant.

3.4 Model Hyper-parameter Settings

In the data preprocessing section, the image size will be Resized. In this experiment, the length and width of the image are both 650, and the batch size is set to 8. The initial learning rate α is 0.0001, and the learning rate is multiplied by 0.1 every 10 epochs during training. The gradient average coefficient β_1 is set to 0.9, the gradient squared coefficient β_2 is set to 0.999, and ϵ is set to 10^{-8} . This experiment also uses L2 regularization, adding $\lambda \sum_j w_j^2$ to the loss function, and the weight attenuation parameter λ is set to 0.0001. Preventing overfitting during model training plays a role.

Table 1: Performance Comparison of Basic Convolutional Neural Networks.

Model	Top-1/%	Top-5/%	params
VGG-16	80.5%	94.9%	138M
ResNet-50	88.1%	99.7%	25.5M
InceptionV3	89.4%	99.5%	27.4M
MobileNetV2	87.0%	99.6%	3.4M

Note: Top-1: The accuracy rate of the category with the highest predicted probability is the same as the actual result. Top5 is also the same concept.

4 Analysis and Comparison of Experimental Results for Improved Convolutional Neural Network

4.1 Comparison of Basic Model Results

From the top-1 accuracy rate in Table 1, it can be seen that the InceptionV3 model has the best performance in the disease data set, reaching 89.4%, and the Top-5 accuracy rate of ResNet-50 is 99.7%. Of course, MobileNetV2 also has the advantage of small model. Table 1 is the test set accuracy rate from the basic model and using some commonly used deep learning methods, such as data preprocessing, network layer parameter initialization, iteration number, and learning rate change strategy. In 4.2 and

4.3, we will introduce some model structure improvement and training optimization that have a great influence on the accuracy of the model, and compare it with Table 1.

4.2 Improve the Network Structure

4.2.1 ResNet-50 Structure. Model improvement is a minor adjustment to the network results, adjusted to a specific convolutional layer. The adjustments in this section of the experiment have almost no effect on the overall computation of the model, but have a non-negligible effect on the accuracy of the model. This section uses ResNet-50 as an example to study the performance of the improved network on the crop disease dataset.

Here is a brief introduction to ResNet-50. A ResNet-50 network starts from the input as shown in Figure 2. It contains a 7×7 convolution kernel with 64 channels, a stride of 2, followed by a 3×3 maximum pooling layer with a stride of 2. The next four large The module has a downsampling block and two residual

blocks in the fourth module. The downsampling block contains two paths, and the left path contains three convolution kernels, which are 1×1 , 3×3 , and 1×1 , respectively. The first 1×1 convolution kernel has a stride of 2, the rest is 1, and the final output channel is 2048. There is only one 1×1 convolution kernel in the right path, the stride is 2, and the output channel is also 2048. The two path outputs are summed to finally get the output of the downsampled block.

Table 2: Performance Comparison of ResNet-50, ResNet-A, and ResNet-B.

Model	Top-1%	Top-5%	params
ResNet-50	88.1%	99.7%	25.5M
ResNet-A	89.5%	99.8%	25.5M
ResNet-B	89.8%	99.8%	25.5M

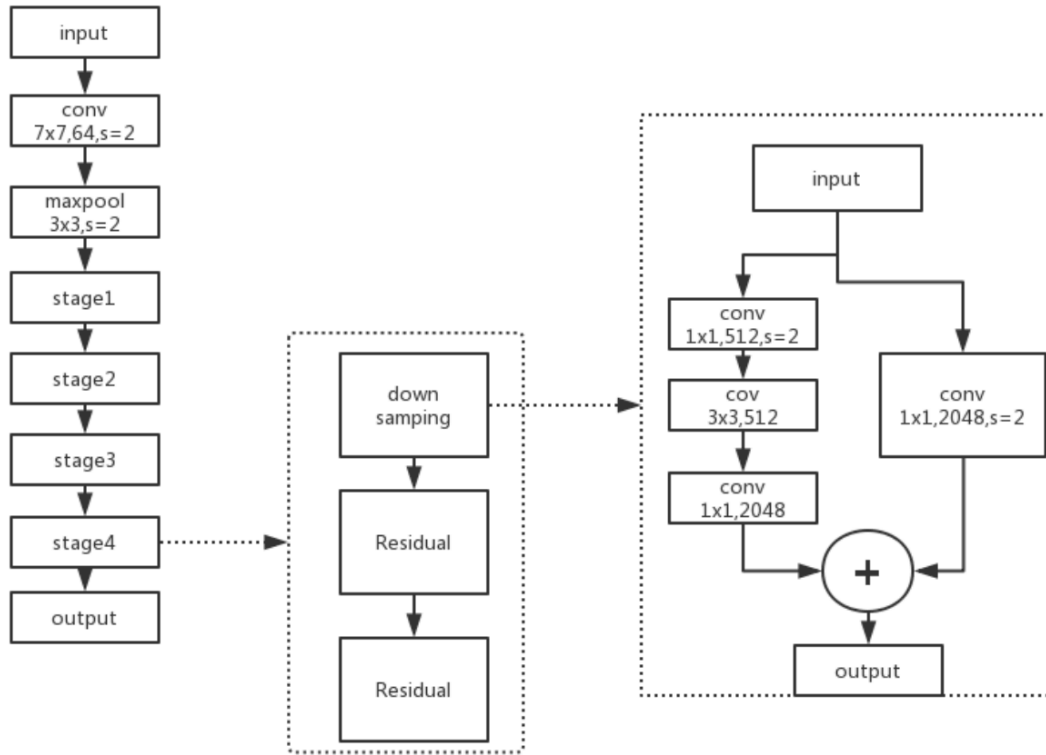


Figure 2: Architecture of ResNet-50.

Table 3: Comparison of Experimental Results after Optimization of Model Labels.

	VGG-16		ResNet-B		InceptionV3		MobileNetV2	
	Top-1%	Top-5%	Top-1%	Top-5%	Top-1%	Top-5%	Top-1%	Top-5%
Reference	80.5%	94.9%	89.8%	99.8%	89.4%	99.5%	87.0%	99.6%
+Label smoothing	81.6%	95.7%	91.2%	99.9%	90.6%	99.8%	87.7%	99.7%

4.2.2 Improving ResNet-50. In this section, two changes are made to the model ResNet-50. The modified network structure is shown in ResNet-A and ResNet-B in Figure 3.

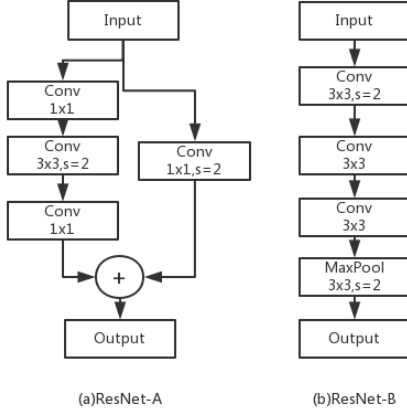


Figure 3: Improved ResNet-50 (Partial Structure).

In ResNet-A, the step size of the first 1×1 convolution kernel is changed to 1, and the step size of the second 3×3 convolution kernel is changed to 2. The reason for this improvement is that when a 1×1 convolution kernel step is used for a convolution operation of 2, about three-quarters of the feature parameters are lost, and the 3×3 convolution kernel step is adjusted to 2 without loss. Characteristic Parameters. The output shape and improvement did not change compared to before the improvement.

The ResNet-B replaces the 7×7 convolution kernel at the beginning of the structure with three 3×3 convolution kernels. Replacing the 7×7 convolution kernel with multiple 3×3 convolution kernels not only has no difference in field of view but also has the following two advantages: 1. The improved parameter amount is reduced. 2. The network has obtained more nonlinear transformations, and the ability to learn features has been enhanced. Table 2 shows the comparison of experimental results.

As can be seen from Table 2, compared with ResNet-50, ResNet-A has an increase of about 1.4% in Top-1% and a 0.1% increase in Top-5%. Compared with ResNet-50, ResNet-B has an increase of about 1.7% for Top-1% and a 0.1% increase for Top-5%. On the other hand, these two improvements do not have a large impact on the amount of model parameters. From the experimental results, these two improvements have a certain degree of improvement in model performance.

4.3 Data Tag Optimization

Generally, our label on training data will be represented by One-hot. When we use the model to fit the data of this One-hot tag, it will bring us some problems: 1. The generalization ability

of the model is poor, and it is easy to cause over-fitting. The one-hot tag encourages the gap between the category of the image and other categories to be as large as possible, not only making the prediction probability of other categories not provide any help to the model, which makes our model too convinced of the predicted category. To improve this approach, using Label Smoothing, the tag expression after label smoothing is as follows:

$$y_i = \begin{cases} 1-\epsilon_1 & i=y \\ \epsilon_1/(K-1) & \text{otherwise} \end{cases} \quad (11)$$

In Equation (11), take 0.1 in this experiment, K is the number of dataset categories, so take 59. Table 3 shows the comparison of experimental results.

As can be seen from the Top-1% of the Table 3, VGG-16, ResNet-B, InceptionV3, and MobileNetV2 increased by 1.1%, 1.4%, 1.2%, and 0.7%. Top-5% also increased by 0.8%, 0.1%, 0.3% and 0.1%. The performance of ResNet-B is the best, with Top-1% up to 91.2% and Top-5% having 99.8% accuracy. On the other hand, this crop disease experiment proved that the distribution of One-hot tags would have a negative impact on the disease identification model.

5 Conclusions

- 1) The experimental crop disease data set has 40,772 images. The data set has 59 categories, with 10 crops (*apples, tomatoes, cherries, strawberries, etc.*) and 26 disease samples. There are 23 types of diseases classified into general and severe according to the degree of disease. The performance of VGG-16, ResNet-B, InceptionV3 and MobileNetV2 in crop disease identification was studied. The accuracy of Top-1 was 80.5%, 88.1%, 89.4%, 87.0%. The accuracy of Top-5 was 94.9%, 99.7%, 99.5%, 99.6%.
- 2) In order to improve ResNet-50, this paper replaces a 7×7 convolution kernel with three 3×3 convolution kernels and changes the size of the downsampling convolution kernel step. the top-1 of ResNet-A has a 1.4% improvement, and the top-1 of ResNet-B has a 1.7% improvement. With Lable smoothing, Top-1 accuracy for VGG-16, ResNet-B, InceptionV3, and MobileNetV2 increased by 1.1%, 1.4%, 1.2%, and 0.7%. The final result of the experiment ResNet-B model has the best performance, Top-1 has an accuracy of 91.2%, and Top-5 has an accuracy of 99.9%.
- 3) The combination of deep learning and crop disease identification, the time cost of data collection needs to be considered, and we can study the time cost of data collection in the future. Deep learning identifies different types of diseases and stages of occurrence, which is of great significance for the detection of crop diseases and the prevention of pests and diseases.

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